

Benjamin C. Chu

Website: www.bcchu.com

Address: 7000 Hawaii Kai Drive Apt 2317, Honolulu, HI 96825

Citizenship: USA

Email: bcchu@hawaii.edu

Mobile: (808) 429-8570

Received his PhD in economics from the University of Hawai'i at Mānoa in 2023 with a research focus in health economics, health policy, and applied econometrics. Results-driven professional with 8+ years of experience in policy analysis, academic research, econometrics, machine learning, and statistical analysis. Skilled in various programming languages for data science and adept at drafting and presenting both statistical and legislative reports. Experienced in researching Medicare and Medicaid policies and delivering presentations to diverse audiences, including both technical and non-technical stakeholders. Strong background in designing and delivering university-level courses.

EDUCATION

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- University of Hawai'i at Mānoa Honolulu, Hawai'i
• **Ph.D. - Economics** May 2023
Dissertation: *Essays on Health Coverage Expansions and its Impacts on Low-Income Populations*
Committee: Timothy Halliday, Victoria Fan, Ruben Juarez, Sang-Hyop Lee & Teresa Molina
 - University of Hawai'i at Mānoa Honolulu, Hawai'i
• **M.A. - Economics** May 2019
 - University of Hawai'i at Mānoa Honolulu, Hawai'i
• **B.S. - Family and Consumer Sciences** May 2015

WORK EXPERIENCE

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- Social Science Research Analyst | Centers for Medicare and Medicaid Services (CMS)** Remote
• Supervisor: Elizabeth Goldstein, (410) 786-6665 July 2023 - Present
Address: 7500 Security Boulevard, Baltimore, MD 21244
 - Collaborated in the processing, publishing, and dissemination of the Medicare Part C and D Star Ratings, leveraging data analysis and automation techniques in SAS, SQL, Excel, and Tableau to improve the accuracy and timeliness of reporting.
 - Applied statistical techniques to analyze and present key findings and performance metrics from the Star Ratings program, delivering actionable insights to senior leadership and external partners.
 - Engaged with clients, consultants, and representatives from national health organizations to address inquiries on Star Ratings methodology and regulations, and efficiently processed data requests.
 - Research Assistant | Telecommunications and Social Informatics Program (TASI)** Honolulu, Hawai'i
• Supervisor: Norman Okamura, (808) 956-2909 Oct 2022 - June 2023
Address: 2424 Maile Way, Saunders Hall 704, Honolulu, HI 96822
 - Provided technical, analytic, and coordination support for research activities, reporting, data requests, and presentations; including development of codes, workflow, briefs, and Standard Operation Procedures for various data science and Health Information Technology projects.
 - Assisted with research on various healthcare and healthcare system topics, including conducting literature reviews, writing academic papers, and preparing findings for dissemination to stakeholders and non-technical audiences.
 - Conducted statistical analysis and modeling of health claims data using statistical software such as R, SAS, Python, or other data science tools.
 - Lecturer | Department of Economics, University of Hawai'i at Mānoa** Honolulu, Hawai'i
• Supervisor: Sang-Hyop Lee, (808) 956-7041 Jan 2019 - July 2023
Address: 2424 Maile Way, Saunders Hall 542, Honolulu, HI 96822
 - Taught undergraduate courses in the subjects related to microeconomic theory, macroeconomic theory, econometrics and data analysis and visualization.
 - Responsible for creating and instructing a course in data science as part of a focused economics bridge program between Kapi'olani Community College (KCC) and the University of Hawai'i at Mānoa (UHM) for four COVID-19 Affected groups, Native Hawaiians, Pacific Islanders, Filipinos and women.
 - Designed and constructed a learning curriculum that is easily accessible for students both in classrooms and online environments.

- Research Assistant | Pacific Health Analytics Collaborative (PHAC)** Honolulu, Hawai'i
Supervisor: Victoria Fan, (617) 398-0797 Sep 2019 - Sep 2022
Address: Biomedical Sciences, 1960 East-West Rd, Honolulu, HI 96822
 - Conducted economic research that proposed innovations in Medicaid, risk adjustment systems, and diagnostic classification for the purpose of improving health care quality and accessibility.
 - Applied methods in economics and econometrics to conduct statistical analyses of health policy topics using electronic health records, medical claims, and survey data.
 - Utilized R, Python, and PowerBI to perform tasks in data cleaning, web scraping, quality assurance, statistical inference and data visualization.
 - Prepared summary reports, academic papers, data dashboards, briefings, and data quality assessments to be presented to various community stakeholders, health professionals, and non-technical audiences.
- Educational Coach | Center on Disability Studies, University of Hawai'i at Mānoa** Honolulu, Hawai'i
Supervisor: Sean Nagamatsu, (808) 292-5503 Aug 2018 - Jan 2019
Address: 1410 Lower Campus Road, 171F Honolulu, HI 96822
 - Assisted students with special health care needs in improving their academic and self-management skills such as time management, organization, and study habits.
 - Helped students identify and build strengths to transfer abilities across domains, develop individual plans for academic support, and foster the growth of affective skills.
- Teaching Assistant | Department of Economics, University of Hawai'i at Mānoa** Honolulu, Hawai'i
Supervisor: Sang-Hyop Lee, (808) 956-7041 Aug 2017 - Aug 2019
Address: 2424 Maile Way, Saunders Hall 542, Honolulu, HI 96822
 - Taught weekly sessions in microeconomics and econometrics for undergraduate students.
 - Assisted in the creation and grading of assignments and exams for undergraduate economic courses.

RESEARCH EXPERIENCE

6+ years of experience conducting independent, original economic research in health economics, health policy, and applied econometrics. Details of research-related experience include:

- Analyzing the effects of health programmes connected to Medicare and Medicaid using programming techniques, economic theory, econometrics, and machine learning.
- Performing independent and original research that serves to motivate policy development for Medicare and Medicaid.
- Implementing statistical econometric and machine learning techniques in the analysis and interpretation of economic research questions.
- Preparing a written analysis of statistical findings to be disseminated in legislative reports or submitted to academic journals.
- Presenting research at conferences and meetings with various professional organizations, stakeholders, academic audiences and non-technical audiences.

From Dissertation: *Essays on Health Coverage Expansions and its Impacts on Low-Income Populations*

- Who Did the ACA Medicaid Expansion Impact? Estimating the Probability of Being a Complier (Sole-Authored) (Health Economics)**
 - Used data from the American Community Survey (ACS) to estimate the conditional likelihood of being induced by ACA Medicaid expansion to enroll in Medicaid (i.e., probability of being a complier).
 - Found that in non-expansion states, many of which have high Black populations, the probability of being a complier is higher for Black individuals than for other racial/ethnic groups, suggesting that Black individuals in non-expansion states would be the largest beneficiaries of any new expansions.
 - First study to apply techniques in complier analysis to estimate the compliers from the ACA Medicaid expansion.
- Laying Down the Welcome Mat: Measuring the Impacts of the Medicaid Expansion for Children Previously Eligible for Medicaid (Sole-Authored) (Work in Progress)**
 - Used data from the American Community Survey (ACS) to estimate the impact of the ACA Medicaid expansion for children who were eligible for Medicaid prior to the expansion.
 - Crafted measurements of Medicaid eligibility for children using both Modified Gross Adjustment Income (MAGI) thresholds from the Kaiser Family Foundation (KFF) and poverty thresholds from the Census Bureau.
 - Found modest yet significant increases in public coverage across all years for children who were previously eligible for Medicaid and CHIP prior to the expansion, but with significant crowding out in employer-sponsored insurance.

Other Research:

- **Assessing the Effectiveness of Risk Adjustment Systems in Recognizing Dual Diagnosis and Co-occurring Conditions for Mental Health and Substance Use Disorders (Work in Progress)**
 - Estimating the performance of current risk adjustment models for health insurance marketplaces and Medicare in predicting the risk score that accurately reflects the costs associated with treating those experiencing multiple co-occurring conditions in mental health and substance use.
 - Developing a novel comorbidity index that captures the prevalence of co-occurring mental health conditions and dual diagnoses for high risk enrollees.

PUBLICATIONS

1. **Chu, B. C.** (2023). Who did the ACA medicaid expansion impact? Estimating the probability of being a complier. *Health Economics*.

POLICY PAPERS & LEGISLATIVE REPORTS

1. Medicare 2025 Part C and D Star Ratings Technical Notes. Washington, DC. October 3rd, 2024. Web Version: Report
2. Fact Sheet 2025 Medicare Advantage and Part D Star Ratings. Washington, DC. October 10th, 2024. Web Version: Report
3. Medicare 2024 Part C and D Star Ratings Technical Notes. Washington, DC. March 13th, 2024. Web Version: Report
4. **Chu, B. C.**, Nakano, L., Do, B., Emory, C., Fan, V.Y. Hawai'i State Epidemiologic Report and County Profiles. Honolulu, HI. August 6th, 2022.
5. Aoki, C., Valdez, K., Abe, A., Burke, K., **Chu, B. C.**, Subia, C., Lumbao, S., Untalan, R., McKee, S., Ha, E., Talagi, D., Fontanilla, T., Worley, K., Galanis, D., Bolan, A., Fan, V.Y. Hawai'i Opioid Initiative Evaluation Report 2021. Honolulu, HI. August 31st, 2021.
6. Abe, A., **Chu, B. C.**, Fan, V.Y. State Opioid Response Technical Assistance Plan, 2021.
7. **Chu, B. C.**, Fontanilla, T., Fan, V.Y. University of Hawai'i at Mānoa. Overdose Data to Action Plan (OD2A) Laulima Outpatient Dataset - Data Quality Assessment, 2021.
8. **Chu, B. C.**, Fontanilla, T., Fan V.Y. University of Hawai'i at Mānoa. Overdose Data to Action Plan (OD2A) Laulima Inpatient Dataset - Data Quality Assessment, 2021.

SKILLS SUMMARY

- **Programming Languages:** R, Stata, SAS, Python, SQL, HTML, VBA
- **Integrated Development Environments:** RStudio, SAS Enterprise, Jupyter Notebook, Spyder, MySQL Workbench
- **Operating Systems:** Microsoft Windows OS, Mac OS
- **Media/Design/Data Visualization:** L^AT_EX, Adobe Illustrator, Microsoft PowerPoint, Microsoft Excel, PowerBI, Tableau Desktop, Tableau Prep Builder
- **Research:** Health Economics, Risk Adjustment, Regression Analysis, Machine Learning, Model Evaluation, Statistical Modeling, Research Design, Academic Writing, Qualitative Analysis, Quantitative Analysis, Econometrics

CERTIFICATIONS & SPECIALIZATIONS

- **Google Advanced Data Analytics Professional Certificate** *Coursera, Google*
- **IBM Machine Learn Professional Certificate** *Coursera, IBM*
- **Data Science Fundamentals with Python and SQL Specialization Certificate** *Coursera, IBM*
- **SQL for Data Science Course Certificate** *Coursera, UC Davis*

TEACHING

- **ECON 130: Principles of Microeconomics** (Teaching Assistant) *Fall 2016, Spring 2017, Fall 2018, Spring 2019*
- **ECON 130: Principles of Microeconomics** (Instructor) *Spring 2019, Spring 2020, Fall 2022*
- **ECON 131: Principles of Macroeconomics** (Instructor) *Summer 2021*
- **ECON 256: Data Analysis and Visualization** (Instructor) *Summer 2022, Spring 2023, Summer 2023*
- **ECON 300: Intermediate Macroeconomics** (Instructor) *Summer 2019, Summer 2020*
- **ECON 425: Introduction to Econometrics** (Teaching Assistant) *Spring 2019*
- **ECON 628: Econometrics I** (Teaching Assistant) *Spring 2023*
- **ECON 629: Econometrics II** (Teaching Assistant) *Fall 2022*
- **ECON 674: Health Economics and Policy** (Guest Lecturer) *Fall 2022*

HONORS AND AWARDS

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- Department of Economics Best Student Publication Award, University of Hawai'i at Mānoa 2023
 - College of Social Science Excellence in Teaching Award (Nomination), University of Hawai'i at Mānoa 2023
 - Burnham O. Campbell Scholarship, University of Hawai'i at Mānoa 2022
 - Pi Gamma Mu, University of Hawai'i at Mānoa 2022
 - Frances Davis Graduate Student Teaching Award (Nomination), University of Hawai'i at Mānoa 2017-2019
 - Graduate Assistantship, University of Hawai'i at Mānoa 2017-2019
 - Phi Upsilon Omicron, University of Hawai'i at Mānoa 2012-2015
 - Golden Key International Honor Society, University of Hawai'i at Mānoa 2012-2015
 - Band Scholarship, University of Hawai'i at Mānoa 2010-2016

CONFERENCES

2023:

2. "Who Did the ACA Medicaid Expansion Impact? Estimating the Probability of Being a Complier". *American Society of Health Economists (ASHEcon)*, St. Louis, Missouri, June 11th, 2023.
1. Keynote Speaker. *Special Education Conference I Ke Alu Like*, Honolulu, Hawai'i, March 14th, 2023.

2022:

4. "Who Did the ACA Medicaid Expansion Impact? Estimating the Probability of Being a Complier". *Hawai'i Economic Association Graduate Student Showcase*, Honolulu, Hawai'i, November 30th, 2022.
3. "Find Your Peace, Find Your Space". *Special Parents Information Network Conference*, Honolulu, Hawai'i, October 22nd, 2022 (with Thao Le).
2. "Who Did the ACA Medicaid Expansion Impact? Estimating the Probability of Being a Complier". *University of Hawai'i at Mānoa Applied Microeconomics Workshop*, Honolulu, Hawai'i, September 20th, 2022.
1. "Does Smoking Affect Wages?". *Workshop on the Economics of Risky Behavior*, Honolulu, Hawai'i, July 9th, 2022 (Discussant) (with Gregor Pfeifer).

2021:

1. "Laying Down the Welcome Mat: Measuring the Impacts of the Medicaid Expansion for Children Previously Eligible for Medicaid". *Hawai'i Economic Association Graduate Student Showcase*, Honolulu, Hawai'i, November 4th, 2021.

2020:

1. "Autism Supports See Through My Eyes". *Special Parents Information Network Conference*, Honolulu, Hawai'i, October 17th, 2020 (with Alysha Kim).

2019:

1. "A Horse of Different Color: Strategies to Support Students with Autism Focusing on Motivation". *Special Parents Information Network Conference*, Honolulu, Hawai'i, April 13th, 2019 (with Ann N. Garfinkle).

2018:

1. "Mindful Meditation and Communication Strategies". *Special Parents Information Network Conference*, Honolulu, Hawai'i, April 21st, 2018.

2017:

1. Keynote Address. *Special Parents Information Network Conference*, Honolulu, Hawai'i, April 22nd, 2017.

REFERENCES

Prof. Timothy Halliday
Professor
Department of Economics
University of Hawai'i at Mānoa
☎ (808) 956-8615
✉ halliday@hawaii.edu

Prof. Victoria Fan
Associate Professor
Center on Aging
University of Hawai'i at Mānoa
☎ (617) 398-0797
✉ vfan@hawaii.edu
✉ vfan@post.harvard.edu

Dr. Elizabeth Goldstein
Director
Center for Medicare
Centers for Medicare and Medicaid
☎ (410) 786-6665
✉ Elizabeth.goldstein@cms.hhs.gov